A Conditional GW170817 Constraint on a Vestigial-Susceptibility Speed Law

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For the assumed gravitational-wave speed law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, the conservative symmetric GW170817 cap $|\Delta c/c| \le \tau$, with $\tau = 3 \times 10^{-15}$, is equivalent to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. This finite interval has center $1+\tau^2$, half-width 2τ and width 4τ. It lies strictly inside the adopted comparison range $[0.1, 10]$ and occupies the fraction $4\tau/9.9$ of its width; that fraction is not a probability without a specified prior. The lower and upper comparison endpoints exceed the speed cap by approximately 2.3×10^{14} and 7.2×10^{14}, respectively, while exact luminality occurs at $\chi_{\rm vest} = 1$. We formalize the scalar identities, interval and endpoint results, and four explicit counterexamples to the conjunction of positivity, cap compatibility and sub-half relative deviation. An illustrative frequency correction is an unmatched product ansatz; its conditional placeholder bound supplies no physical detection band. Neither the spin-two second-sound interpretation nor the susceptibility-dependent speed law is derived from a microscopic metric action here. The physical susceptibility normalization and domain, propagating tensor mode, matter reference and controlled transport remain required for a physical application. The finite interval constrains this adopted law under the source-emission assumptions used to infer the GW170817 cap; it excludes neither all vestigial phases nor the emergent-gravity framework.

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I. INTRODUCTION

Composite-tetrad gravity provides a setting for asking whether metric order can precede a nonzero tetrad expectation. Volovik's vestigial-phase proposal describes precisely such an expectation-valued metric [2]. A separate de Sitter two-fluid calculation gives a collective second-sound speed $s_2 = c$ [3]. That work discusses possible graviton interpretations and leaves the type of the collective mode open; it does not derive the spin-two identification adopted here.

We assume both a spin-two interpretation of a candidate metric-channel excitation and the kinematic law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, where $\chi_{\rm vest}$ is a free, dimensionless normalized susceptibility. The RPA resummation supplies a conditional response in terms of a bare susceptibility and coupling, not a numerical physical value or a derivation of this speed law. A quadratic action, constrained propagator, tensor polarizations and matter coupling are still required.

The August 17, 2017 binary-neutron-star merger (GW170817) [1] placed an extraordinarily tight bound on $c_{\rm GW}$: by comparing the gravitational-wave arrival time at LIGO/Virgo against the gamma-ray-burst signal at Fermi/INTEGRAL, the multi-messenger collaboration constrained $-3\times10^{-15} \le (c_{\rm GW}-c)/c \le +7\times10^{-16}$. We adopt the conservative symmetric envelope $|\Delta c/c| \le 3 \times 10^{-15}$ under the collaboration's distance and source-emission assumptions. Alternative emission histories can weaken the inferred propagation constraint.

We formalize the consequences of the assumed speed law and compare them with the conservative GW170817 envelope. For $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, compatibility requires a sub-interval of the adopted range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by ~ 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width exactly 4τ in $\chi_{\rm vest}$ and lies strictly interior to $[0.1, 10]$, containing the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9$; its probability mass depends on the prior.

This is a conditional constraint on one specific kinematic identification (second-sound mode = graviton). It excludes the two natural-range endpoints under the assumed speed law while allowing the finite interval above. The width comparison alone neither refutes emergent gravity or the ADW program nor tests the linearized-Einstein identification [11]. Possible microscopic mechanisms are discussed in Sec. VII.

II. SETUP

A. Vestigial-susceptibility identification

The RPA resummation used in `VestigialSusceptibility.lean` has the conditional closed form $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$. The bare susceptibility χ_0 and effective coupling $\gamma_\star$ are inputs; this formula does not evaluate them or supply a normalized physical $\chi_{\rm vest}$. The comparison interval $\chi_{\rm vest} \in [0.1, 10]$ spans one decade on either side of unity and is adopted, not calculated. It is dimensionless. Deriving its physical normalization and stable parameter domain requires the metric-channel source, projector, regulator and microscopic response. Vergeles's unitarity analysis [5] does not calculate that susceptibility or interval.

B. The assumed $c_{\rm GW}$ formula

For a spin-two metric-channel interpretation we adopt

$$c_{\rm GW}^2(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}. \quad (1)$$

The natural anchor $\chi_{\rm vest} = 1$ recovers $c_{\rm GW} = c$ exactly. The deviation

$$\Delta c/c(\chi_{\rm vest}) = \sqrt{\chi_{\rm vest}} - 1 \quad (2)$$

is dimensionless, vanishes iff $\chi_{\rm vest} = 1$, and is strictly monotone in $\chi_{\rm vest}$ on $[0, \infty)$.

In Lean (`GravitationalWaves.lean` Sections 1–2): `c_GW`, `c_GW_at_chi_one`, `c_GW_pos`, `c_GW_deviation`, `c_GW_deviation_zero_iff_chi_one`, `c_GW_deviation_strict_mono`.

C. The GW170817 correctness-push

The GW170817 [1] bound is $|\Delta c/c| \le \tau$ with $\tau = 3 \times 10^{-15}$. We define the predicate $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \le \tau$ and state the correctness-push biconditional:

$$\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]. \quad (3)$$

Theorem A (`c_GW_match_iff_chi_close_to_one`). For $\chi_{\rm vest} \ge 0$, Eq. (3) holds. The susceptibility interval has center $1 + \tau^2$ and half-width 2τ, since its endpoints are $1 \mp 2\tau + \tau^2$. Exact luminality holds only at $\chi_{\rm vest} = 1$; finite-tolerance compatibility holds throughout this interval.

The proof is a $\sqrt{\cdot}$/squaring round-trip via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, with a nonlinear-arithmetic step closed by `nlinarith` using $\sqrt{\chi_{\rm vest}}^2 = \chi_{\rm vest}$ on the nonnegative domain.

III. THE NATURAL-RANGE CONSTRAINT

The GW170817-compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width exactly $4\tau = 1.2 \times 10^{-14}$ in $\chi_{\rm vest}$ (`ligo_window_width_eq`). The natural range $[0.1, 10]$ has width 9.9 (`natural_range_width_eq`). The compatible window is not disjoint from the natural range: it is nested strictly inside it (`ligo_window_subset_natural_range`) and the two meet at $\chi_{\rm vest} = 1$ (`ligo_window_inter_natural_range_nonempty`).
The content of the constraint is the width ratio: scaling the window up by 10^{14} still leaves it narrower than the natural range (`ligo_window_width_lt_natural_range_width_by_1e14`).

We make the exclusion of the extremes sharp with two endpoint-falsifier theorems.

Theorem B (`natural_lower_violates_ligo`). At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and `LigoSatisfied` fails.

Theorem C (`natural_upper_violates_ligo`). At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and `LigoSatisfied` fails.

The proofs use $\sqrt{1/10} \le \sqrt{1/4} = 1/2$ and $\sqrt{10} \ge \sqrt{4} = 2$, both via `Real.sqrt_le_sqrt` and `Real.sqrt_sq`, closed by `linarith`. The Lean source supplies proofs of these two concrete real-number inequalities.

Bundled Theorem D (`vestigial_natural_range_violates_ligo`). Both endpoints of the natural range fail `LigoSatisfied`.

IV. TRACKED HYPOTHESIS: VESTIGIAL MODE = GRAVITON

We collect the scalar compatibility conditions into a Lean proposition and prove explicit pointwise counterexamples. These conditions describe the adopted model, not a microscopic identification.

A. Endpoint exclusion and its physical scope

The endpoint theorem `natural_endpoints_violate_ligo_window` excludes the adopted values $\chi_{\rm vest} \in \{0.1, 10\}$ under the assumed speed law. The reference-speed-independent form `natural_endpoints_violate_ligo_for_every_c` rests on `c_GW_relative_deviation_eq` and holds for every $c \ne 0$, including the physical domain $c > 0$. These are endpoint constraints, not a derivation or impossibility proof of a microscopic graviton. A metric action, physical state, response normalization, tensor propagator and matter reference remain necessary to apply the scalar model to a physical mode.

Both are statements about the two endpoints, and neither excludes interior values: $\chi_{\rm vest} = 1$ satisfies the cap (`ligo_satisfied_at_chi_one`). Together with the width ratio of Sec. III, they place the identification's surviving parameter space more than 14 orders of magnitude below the natural range without excluding that range.

B. The bundled hypothesis

The recorded Prop has three conjuncts: positive susceptibility (P1), GW170817 compatibility (P2), and sub-half-deviation (P3'):

$$\begin{aligned}&\texttt{H_VestigialModeIsGraviton}(\chi_{\rm vest}) \\ &\quad \equiv \chi_{\rm vest} > 0 \\ &\qquad \wedge \texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \\ &\qquad \wedge |\Delta c/c(\chi_{\rm vest})| < 1/2.\end{aligned}$$

At the fixed tolerance $\tau = 3 \times 10^{-15}$, P2 implies P3' because $\tau < 1/2$. On the nonnegative physical domain, the compatible interval in Eq. (3) also supplies P1. The three clauses are therefore not independent; the recorded Prop retains all three clauses.

Theorem E (`H_VestigialModeIsGraviton_at_one`). $\chi_{\rm vest} = 1$ discharges the bundle.

Falsifier theorems (four pointwise failures). `H_VestigialModeIsGraviton_fails_at_natural_lower` (P2 fails), `H_VestigialModeIsGraviton_fails_at_natural_upper` (P2 fails), `H_VestigialModeIsGraviton_fails_at_zero` (P1 fails), `H_VestigialModeIsGraviton_fails_at_quarter` (P3' fails: at $\chi_{\rm vest} = 1/4$, $\Delta c/c = -1/2$ saturates the strict P3' bound and also fails P2). The quarter-point witness satisfies P1 and shows that P1 alone does not imply P3'; it does not establish independence from P2.

The falsifiers show that the Prop is not universally satisfied; Theorem E supplies a satisfying point. On the nonnegative physical domain its compatible set is precisely the interval in Eq. (3), of width $4\tau = 1.2 \times 10^{-14}$ and containing $\chi_{\rm vest} = 1$. Its probability mass depends on the prior; under a uniform prior on $[0.1, 10]$ it equals $4\tau/9.9$.

V. ILLUSTRATIVE DISPERSION CORRECTION

We consider the real-product ansatz

$$\delta_{\rm disp}(\omega) = \gamma\omega. \tag{4}$$

This is an adopted algebraic correction, not a derived leading term of a microscopic derivative expansion. Identifying it with a fractional frequency shift or a relative propagation-speed correction requires a physical dispersion relation.

The proposed $\gamma = \Gamma_H/c_{\rm GW}^2$ remains unmatched. The definition `SecondOrderSK.GammaH` $= (\gamma_1 + \gamma_2)(\kappa/c_s)^2$ uses acoustic transport coefficients γ_1, γ_2, an inverse-time horizon scale κ and sound speed c_s. It supplies an acoustic horizon damping parameter. Its notation does not identify it with the metric-channel coefficient required here. A physical application requires source/state matching and a frequency and unit convention. If ω denotes a physical frequency, γ must carry inverse-frequency units for their product to be dimensionless. The real-valued formalization assigns no units or angular-to-ordinary-frequency conversion. Under the acoustic SI assignments $[\Gamma_H] = \mathrm{s}^{-1}$ and $[c_{\rm GW}] = \mathrm{m/s}$, the proposed ratio has units $[\Gamma_H/c_{\rm GW}^2] = \mathrm{s/m}^2$, so it does not yet supply the required inverse-frequency coefficient.

The definition `GravitationalWaves.dispersion_correction` is the product $\gamma\omega$. For real coefficients a, b, its zero-coefficient, linearity and absolute-bound theorems establish

$$\delta_{\rm disp}(0, \omega) = 0, \tag{5}$$

$$\delta_{\rm disp}(a + b, \omega) = \delta_{\rm disp}(a, \omega) + \delta_{\rm disp}(b, \omega), \tag{6}$$

$$|\delta_{\rm disp}(\gamma, \omega)| \le |\gamma|\,|\omega|. \tag{7}$$

For a positive numerical probe ω, `dispersion_within_ligo_iff` gives the correction-only criterion

$$|\gamma\omega| \le \tau \iff |\gamma| \le \tau/\omega. \tag{8}$$

At the numerical probe $\omega = 100$, the assumed placeholder $|\gamma| \le 10^{-30}$ gives $|\delta_{\rm disp}| \le B = 10^{-28} < \tau$. The corresponding placeholder theorem identifies neither a physical inspiral frequency nor a measured coefficient. This calculation supplies no physical frequency-band bound.

Under the further additive assumption $\delta_{\rm tot} = \delta_0 + \delta_{\rm disp}$, where $\delta_0 = \sqrt{\chi_{\rm vest}} - 1$, a specified signed correction is compatible exactly when $|\delta_0 + \gamma\omega| \le \tau$. If only $|\delta_{\rm disp}| \le B$ is known, the reverse triangle inequality gives $|\delta_{\rm tot}| \ge |\delta_0| - B$. Thus $|\delta_0| > \tau + B$ excludes restoration by every correction within that bound. For a near-boundary offset $\delta_0 = s(\tau + \varepsilon)$ with $s \in \{-1, 1\}$ and $0 < \varepsilon \le B$, the permitted choice $\delta_{\rm disp} = -s\varepsilon$ instead gives $\delta_{\rm tot} = s\tau$. Such offsets can be restored within this freely signed additive model; microscopic dynamics have not been shown to supply that correction.

The physical coefficient, sign, normalization and relation to the propagation observable remain to be derived. Product identities and placeholder arithmetic discharge none of these requirements.

VI. RESULT FIGURE

Figure 1 shows the deviation function $\Delta c/c(\chi_{\rm vest})$ over $\chi_{\rm vest} \in [10^{-2}, 10^2]$ on log-$\chi_{\rm vest}$ axis with the adopted range $[0.1, 10]$ highlighted. The GW170817 amber band is unresolved around $\Delta c/c = 0$ at this scale. The plotted endpoint offsets are of order unity; the more-than-fourteen-order width comparison follows from the exact interval algebra, not pixel resolution.

VII. DISCUSSION

A. What this rules out

Under the assumed leading-order law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$, the conservative symmetric cap requires $\chi_{\rm vest} \in [(1 - \tau)^2, (1 + \tau)^2]$. The susceptibility offsets from unity are $-2\tau + \tau^2$ and $2\tau + \tau^2$, approximately $\pm 6 \times 10^{-15}$; the speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but finite measurement tolerance permits the whole nonzero-width interval. Its small relative width within $[0.1, 10]$ alone neither establishes a prior probability nor rules out generic UV theories or requires an exact fixed point at unity.

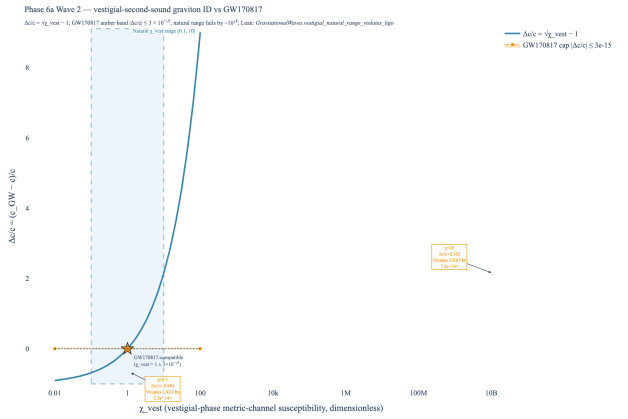


FIG. 1. Conditional speed-law constraint: $\Delta c/c$ deviation across the adopted $\chi_{\rm vest}$ range, versus the GW170817 multi-messenger cap. The amber band $|\Delta c/c| \le 3\times10^{-15}$ is essentially invisible at this scale. The adopted range $[0.1, 10]$ produces $\Delta c/c \in [-0.68, +2.16]$; both endpoints fail GW170817 by $\sim 2.3\times10^{14}$ and $\sim 7.2\times10^{14}$, respectively. The GW170817-compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has finite width 4τ, unresolved at this scale. The gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. Lean: `vestigial_natural_range_violates_ligo`.

B. Open physical constructions

A microscopic mechanism could constrain the metric-channel response through a symmetry, identify a distinct propagating mode, or alter the assumed frequency-independent speed law. The finite-tolerance interval derives none of these possibilities and gives no reason to prefer one. A symmetry selecting $\chi_{\rm vest} = 1$ would suffice, but observational compatibility requires only the finite interval. The illustrative correction in Sec. V remains unmatched to a physical response; its placeholder cannot exclude compensating corrections or establish a controlled frequency band.

C. Comparison with the induced Newton coupling

The companion study [11] parameterizes an induced Newton coupling by $G_N^{\rm emerg} = \alpha_{\rm ADW}\, 12\pi/(N_f\, \Lambda_{\rm UV}^2)$. Here N_f is the companion's fermion-species count parameter. In units $\hbar = c = 1$, write $M_P = (G_{\rm N,obs})^{-1/2}$. At the adopted cutoff $\Lambda_{\rm UV} = M_P$, exact matching requires $\alpha^\star_{\rm ADW} = N_f/(12\pi)$. Its stated count choices give rounded matching values in $[0.40, 1.27]$, inside the adopted $[0.1, 10]$ comparison interval. The kernel's Dirac-count convention and these Weyl-count choices require reconciliation before this supplies a physical Standard Model normalization. The matching locus is conditional algebra; it does not extract the microscopic coefficient selecting it. That coefficient and $\chi_{\rm vest}$ describe different response normalizations. Neither adopted interval derives its physical value.

VIII. CONCLUSIONS

The scalar model $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ gives an exact, finite GW170817-compatible interval under the adopted conservative cap. Its center, width, endpoint exclusions and prior dependence are established mathematical consequences of the assumed law. The formal tracked conjunction also has explicit counterexamples; its scalar predicates do not construct a physical graviton. The real-product frequency correction supplies conditional algebra, not a physical transport coefficient or a measured band.

A physical realization requires the microscopic metric-channel source and normalized susceptibility, a positive physical state, a constrained propagating tensor mode relative to the same matter/light reference, and controlled transport with units and error bounds. Neither endpoint exclusion nor the narrow width proves those constructions impossible. The comparison range $[0.1, 10]$ is adopted in this calculation; deriving it remains a distinct scientific obligation. No whole vestigial-branch exclusion follows.

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